

Prayas JEE 2025

Maths

DPP: 1

Sequence and Series

Q1 If $\log_5 2, \log_5(2^x + 1)$ and $\log_5(2^x + 5/2)$ are in A.P. then x is equal to

- (A) 1 (B) 2
(C) 3 (D) 4

Q2 If 100 times the 100^{th} terms of an A.P. with non-zero common difference equals the 50 times its 50^{th} term, then the 150^{th} term of this A.P. is

- (A) 150
(B) zero
(C) -150
(D) 150 times its 50^{th} term

Q3 If the 19^{th} term of a non-zero A.P. is zero then the ratio (49^{th} term) : (29^{th} term) is

- (A) 4 : 1 (B) 1 : 3
(C) 3 : 1 (D) 2 : 1

Q4 The sum of three numbers in A.P. is 27 and the sum of their square is 293. Then numbers are

- (A) 5, 10, 15
(B) 4, 9, 14
(C) 3, 8, 13
(D) 6, 11, 16

Q5 If $a_1, a_2, a_3, \dots$ are in A.P. where $a_2 = 76$ and $a_{12} = 26$ then least positive integer n for which $a_n < 0$ is _____

Q6 The 19^{th} term from end of progression 3, 6, 9, 12, ..., 72 is _____.

Q7

The sum of 31 terms of AP $a_1, a_2, a_3, \dots, a_{31}$ is $a_1 + a_7 + a_{13} + a_{19} + a_{25} + a_{31} = 432$ is

- (A) 144 (B) 2160
(C) 2232 (D) 864

Q8 If $a_1, a_2, a_3, \dots$ are in AP and $a_1 + a_7 + a_{16} = 40$ then $S_{15} =$

- (A) 100 (B) 150
(C) 200 (D) 250

Q9 If $\log x, \log \sqrt{6-2x}, \log(x-1)$ are in AP then the number of possible values of x is/are

- (A) 0 (B) 1
(C) 2 (D) 3

Q10 Let $a_1, a_2, a_3, \dots$ be term of AP. If $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + \dots + a_q} = \frac{p^2}{q^2}, p \neq q$ then $\frac{a_6}{a_{21}} =$

- (A) $\frac{41}{11}$
(B) $\frac{7}{2}$
(C) $2/7$
(D) $\frac{11}{41}$

Q11 Three numbers are in A.P. such that their sum is 18 and sum of their square is 158. The greatest among them is

- (A) 10 (B) 11
(C) 12 (D) 14

Q12 If the 9^{th} term of a AP be zero, then the ratio of its 29^{th} and 19^{th} term is

- (A) 1 : 2
(B) 2 : 1
(C) 1 : 3
(D) 3 : 1



Answer Key

Q1 (A)
Q2 (B)
Q3 (C)
Q4 (B)
Q5 18
Q6 18

Q7 (C)
Q8 (C)
Q9 (B)
Q10 (D)
Q11 (B)
Q12 (B)



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